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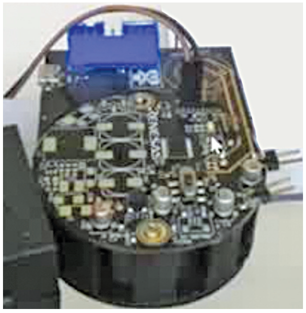
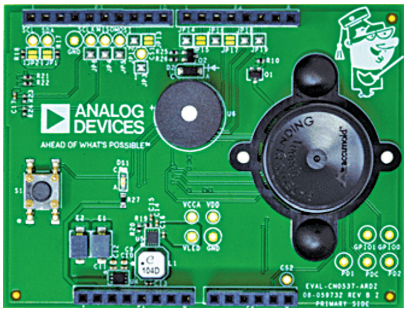
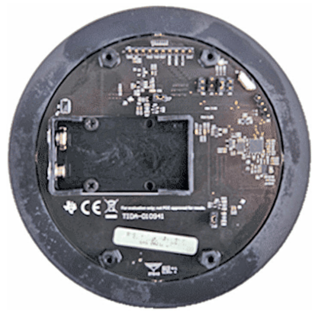
Photoelectric Smoke Detector	Multistandard Verified Smoke Detector	Smoke Detection with Ambient Light Cancellation
		
This reference design for a photoelectric smoke detector features battery monitoring, a dual LED driver, adjustable gain amplifiers, and supports a range of batteries, making it a solution for safety applications.	The reference design for multistandard verified smoke detector features a dual-wavelength optical module with integrated LEDs and photodetectors, streamlining design with pre-verified algorithms and adaptability for various standards.	The design provides a solution for smoke detection, complying with UL217 9th edition standards. It uses a photoelectric system and dual-wavelength optical approach for light rejection and signal-to-noise ratio, reducing false alarms and supporting air quality monitoring.
The RAA239101 reference design by Renesas Electronics Corporation is an analogue front-end (AFE) integrated circuit (IC) designed for low-power consumption and serves as the core component in developing comprehensive smoke detection systems. It is used in conjunction with a microcontroller, a photoelectric emitter/detector setup, a horn, and a few external components to constitute a full-fledged smoke detector solution. Notably, it has a battery monitoring capability to trigger an alarm upon detecting low battery levels and a dual LED driver that facilitates the pulsing of smoke-detection LED emitters with an adjustable current via a digital-to-analogue converter (DAC). It also features two photodiode receiver channels with adjustable gain amplifiers (AGAs), controlled by an analogue-to-digital converter (ADC), to accurately sense smoke particles through the scattering of LED light in a detection chamber. With ultra-low power consumption, the device supports both 9V batteries and a range of 3V to 5V batteries. It includes an LDO for microcontroller power supply, a 10-bit ADC for comprehensive voltage measurement across seven analogue pins, and the capacity to drive two LED emitters with 8-bit current control from 45mA to 600mA.	The CN0537 reference design by Analog Devices is a smoke detector developed to comply with UL 217/UL 268 and similar smoke and fire detection standards. It is compatible with the Arduino form factor, aiming to expedite prototyping and evaluation of embedded smoke detection algorithms. The design includes an algorithm and a dataset based on the ADPD188BI optical module, which is a comprehensive photometric system tailored for smoke detection applications. The optical module employs a dual-wavelength technique, featuring two integrated LEDs that emit light at different wavelengths, and two integrated photodetectors that capture scattered light, enabling the detection of smoke particles. Verified algorithms accelerate the deployment of the smoke detection solution, and additional data can be collected and integrated into the algorithm to adapt it for other standards. Incorporating the optical module instead of traditional discrete smoke detector circuits significantly streamlines the design process, as the optoelectronics and the analogue front end (AFE) are already integrated into the module.	The TIDA-010941 reference design by Texas Instruments offers a cost-effective solution for smoke detection, complying with UL217 9th edition standards through a low BoM approach. The design achieves a high signal-to-noise ratio, enabling advanced algorithms that minimise false alarms and enable precise particulate measurement for air quality monitoring. The device features a sophisticated sensing signal chain with modulation for enhanced ambient light rejection and a compressive sensing method for efficient power use. Its dual-wavelength and angle optical system significantly reduces false alarms and supports air quality assessment. With a signal chain SNR of 77dB and a total current consumption of only 5.8μA, the design ensures exceptional detection clarity and over 10 years of battery life. This design surpasses traditional DC-based systems by offering superior ambient light rejection and SNR, eliminating the need for an optical chamber, and reducing materials and assembly costs. Its dual-wavelength strategy and multi-angle light scattering measurement technique enhance detection accuracy by precisely distinguishing between smoke particles and other particulates, highlighting its potential to improve safety and environmental monitoring in various settings.
The reference design is suitable for household and commercial buildings.	The reference design is particularly well-suited for smoke detection applications.	The reference design finds applications in industrial devices such as air-conditioner units, air purifiers, and air quality sensors, as well as in commercial and residential smoke and heat detectors.
Renesas Electronics Corporation	Analog Devices (ADI)	Texas Instruments (TI)
https://www.electronicsforu.com/electronics-projects/reference-design-for-a-photoelectric-smoke-detector	https://www.electronicsforu.com/electronics-projects/multistandard-verified-smoke-detector-reference-design	https://www.electronicsforu.com/electronics-projects/ambient-light-cancelling-smart-analog-smoke-sensor-interface-reference-design